

BFU is not a three-space property

Source question: arXiv:0804.2832

Supporting theorem: arXiv:1612.05948

Status. Full negative answer implied by later literature; no new mathematical result is claimed here.

Source question. Burshteyn defines a Banach space X to be BFU when every norm-bounded subset of X^* is Fréchet–Urysohn in $\sigma(X^*, X)$. In Section 6, “Open Problems,” p. 89, he asks whether this is a three-space property: if a closed subspace Y and X/Y are BFU, must X be BFU?

This is the standard property that X has a weak-star angelic dual ball.

Counterexample from a later theorem. Martínez-Cervantes proves in Theorem 4 of *Banach spaces with weak*-sequential dual ball* that the Johnson–Lindenstrauss space JL_2 has weak-star sequential order exactly 2. The proof records both

$$JL_2/c_0 \cong \ell_2(\Gamma) \quad \text{and} \quad JL_2 \text{ does not have weak-star angelic dual.}$$

Consequently the exact sequence

$$0 \longrightarrow c_0 \longrightarrow JL_2 \longrightarrow \ell_2(\Gamma) \longrightarrow 0$$

has BFU kernel and quotient: c_0 is separable, and $\ell_2(\Gamma)$ is reflexive. Its middle space is not BFU. Hence the BFU property is not a three-space property.

Provenance. The supporting paper does not cite or name the 2008 question. The answer is therefore a direct identification of a later theorem with the source problem, rather than an explicit literature answer.